



Cooperation in Evolved Modular Soft Robots

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Abstract. Self-assembly is the ability of natural beings to aggregate in a suitable structure in order to tackle complex tasks that could not be solved by single individuals. Robotics tries to replicate this ability to solve complex problems using artificial embodied physical agents. Nevertheless, the potential of self-assembly in robotics has still to be fully exploited. Here we propose self-assembly with voxel-based soft robots, a type of modular soft robots. The goal is to evolve the controller of the robots, inducing the formation of a single structure to move beyond a hole in the terrain that could not be crossed by individual robots. We perform experiments in simulation in increasingly complex arenas. The results show that evolved robots are capable of forming a connected chain and overall succeed in their mission. We also discuss limitations and improvements of the proposed approach.

Keywords: Self-assembly · Soft robotics · Modular robotics · Evolutionary robotics · Neural networks

1 Introduction

The ability to cooperate is a foundational capability for many biological species. When single individuals are not enough the group intervenes, cooperatively facing difficult situations. This allows tackling problems otherwise unbearable, or for which a single individual would spend a great amount of effort. Among different types of cooperation, here we focus on *self-assembly*, which in fact can be found at any scale in nature [18]. In general, self-assembly can be defined as the property of a group of separate entities to autonomously cooperate and come together to form patterns and structures. Many examples are available in nature, notably, the creation of columns and bridges in ant colonies to cross gaps [13].

Due to its importance in tackling difficult tasks, the ability to self-assemble started to be explored also in artificial collective systems. One of the first multi-robot systems able to self-assemble was proposed by O’Grady et al. [12] with the aim of moving over rough terrains. Tuci et al. [17] extended the work by controlling the robots with an artificial neural network (ANN). Additionally, they

tested the ability of collective transportation of heavy objects to a nest, focusing on the design of mechanisms for movement coordination. Recently, Recently, Swissler and Rubenstein [15] have proposed the creation of robots able to form non-lattice structures (*i.e.*, with omnidirectional connections) with the aim to climb on each other and self-assemble into bridges. A recent and detailed survey on the state of the art in robotic self-assembly was provided by Bray and Groß [2]. Here we explore the idea of self-assembly applied to voxel-based soft robots (VSRs) [5, 7], also known as virtual soft robots [11]. VSRs differ from classical robots in that their body is made of a soft material, allowing for more degrees of freedom in movement [3]. This simplifies the problem of coordinated movement, as each VSR can deform to accompany others movements. Additionally, soft-robots reduce the risk of working with humans, minimizing the effect of crashes [1, 6].

We propose evolving the controller of VSRs to obtain a self-assembling behavior in order to complete a task requiring collaboration. Specifically, we consider the case of robots that have to pass over a hole in the ground. The long-term vision is that of soft-robots able to cooperatively navigate complex environments, face a variety of tasks, and safely interact with humans.

2 Experimental Setting

We set up an experiment in which a group of VSRs has to cooperate to pass over a hole in the ground. All the VSRs are identical square-shaped voxel, controlled by a copy of the same controller. The core of the controller is an ANN generated by an evolutionary algorithm (EA). We investigate the performance of controllers produced employing three different fitness functions (FFs). To avoid bias in the results, we run the EA on different arenas characterized by the size of the hole and its distance from the VSRs. Additionally, we investigate how the number of VSRs composing the group affects the success ratio in the experiment.

Each VSR is equipped with 18 sensors, which allow it to perceive the environment and other VSRs [16]: a rotation-angle sensor, a area-ratio sensor that senses the ratio between the current area of the voxel and the rest area (of $1\text{ m} \times 1\text{ m}$); a contact sensor which returns 1 if there is at least one contact and -1 otherwise; a proximity sensor, on a single side, sensing the distance to the closest object up to a range $d = 5\text{ m}$ along a direction normal to the side; two speed sensors in the two orthogonal directions with respect to the current orientation of the voxel; four attachment sensors, one for each side, sensing 1 if the side is attached to another VSR and -1 otherwise; four side-compression sensors, one for each side, sensing the ratio between the current side length and the rest length (of 1 m); four directional-communication sensors, one for each side, each reading communication signals (*i.e.*, scalar numbers) sent by other VSRs within a given range. Additionally, each VSR is endowed with an endogenous signal periodically oscillating and acting as a clock. All sensor output values are normalized in $[-1, 1]$. We choose this configuration for the sensors of the VSRs based on previous experience and studies [14].

Each VSR acts by means of 12 actuators. Four springs, one per side, enable motion by changing their rest value. Four attach/detach actuators, one per side, control aggregation with other VSRs. When an actuation value exceeds a positive threshold τ for a given side and another VSR is close enough in that direction, the two VSRs attract each other (eventually physically attaching). Conversely, when the value exceeds $-\tau$ and the side is attached to another VSR, it detaches from it. Four communication systems, one per side, allow sending scalar signals along a direction normal to the side.

The sensory and the endogenous signals are fed to an ANN controller that produces the control and communication signals. Therefore, the ANN must have 19 input nodes and 12 output nodes. We set up a fully connected feed-forward ANN with 3 hidden layers with 16, 12, and 12 neurons respectively. Including the bias, this results in a total of 836 weights, with tanh as activation function in the neurons. We apply the ANN not at every simulation step, *i.e.*, at a frequency of 60 Hz, but less frequently, at 10 Hz: this way, we avoid vibrating behaviors that would not work in real life [9]. We optimize the ANN weights using an EA, namely covariance matrix adaptation evolution strategy (CMA-ES) [4]. We use the same ANN in every VSR during a simulation. In CMA-ES, we set the population size to 24 and run it for 417 iterations¹.

At the beginning of each simulation, we place the VSRs at fixed positions on the ground, on the left of the hole (see Fig. 1a). The evaluation is based on a FF which computes a fitness value based on the horizontal x -position reached by the VSRs with respect to the start of the arena (the x -axis is oriented left to right). We employ three different FFs, namely: (a) Average group x -position $\mathcal{AVG}$; (b) Maximum VSR x -position $\mathcal{MAX}$; (c) Minimum VSR x -position $\mathcal{MIN}$. $\mathcal{AVG}$ computes the average of the group x -position (see Eq. (1)). $\mathcal{MAX}$ considers the right border x -position of the rightmost VSR of the group (see Eq. (2)). $\mathcal{MIN}$ considers the left border x -position of the leftmost VSR of the group (see Eq. (3)). The result is that $\mathcal{AVG}$ requires most of the group to move right, or a subgroup to move much farther; $\mathcal{MAX}$ just needs one VSR performing well; $\mathcal{MIN}$ requires the entire group to perform well. As the hole might trap the VSRs, a fitness computed with any of the FFs with a value greater than the rightmost boundary of the hole denotes a situation in which some robots (if not all) successfully traversed it. This can happen only if the VSRs successfully self-assemble or, more broadly, if they cooperate. Indeed, the hole is larger than a single VSR and deeper than the height of the piled group, requiring cooperation to pass it. Note that the VSRs before the hole are not guaranteed to know how to self-assemble. They may indeed wander randomly without aggregating.

¹ This value depends on the maximum number of evaluations that we allow: 10'000.

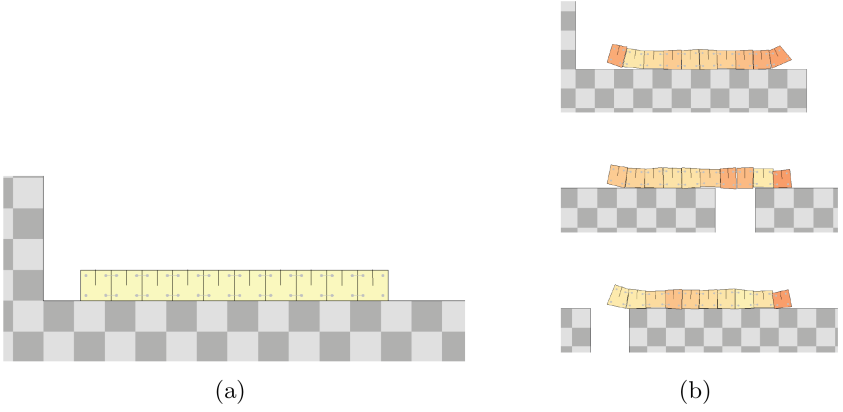


Fig. 1. In (a), 10 VSRs at the beginning of the simulation; the hole is 20m distant from the position of the leftmost VSR. In (b), the VSRs group manages to overcome the hole of type w_2 (i.e., a hole larger than 2 VSRs).

The formulas of the FFs are the following:

$$f_{AVG}(\) = \frac{1}{n} \sum_{i=1}^n x_{c,i} \quad (1)$$

$$f_{MAX}(\) = \max(\{x_{r,i} | i \in [1, n]\}) \quad (2)$$

$$f_{MIN}(\) = \min(\{x_{l,i} | i \in [1, n]\}) \quad (3)$$

where:

- n is the number of VSRs composing the group;
- i is the index of the VSR in the group;
- x_c , x_l , x_r are respectively the center x -position of the VSR, and the left and right border x -position of the VSR at the end of the evaluation.

We perform the experiments in six different arenas characterized by a flat ground with a hole. Half of the arenas present a hole at 12 m distance from the starting VSRs position, while in the other half the hole is at 20 m. The hole can assume three different widths: $w_1 = l + \varepsilon$, $w_2 = 2l + \varepsilon$, and $w_3 = 3l + \varepsilon$, with $l = 1$ m being the voxel side length (see Fig. 1). In other words, the first arena contains a hole slightly larger than one VSR, the second contains a hole slightly larger than two VSRs, and the third contains a hole slightly larger than three VSRs. The goal is to verify the ability to self-assemble in conditions of increasing complexity. We test groups of different sizes: 4, 6, 8, and 10 VSRs. The simulation time of each group of VSRs is limited to 30 s regardless of the arena used. In this time the group is required to pass over the hole and move as far as possible to the right. We perform 30 runs (or replicas) of CMA-ES for each arena. We use JGEA [10] for running the EA and 2D-VSR-Sim [8] for running the simulations.

Table 1. Number of replicas with a performance sufficient to be considered a “success”, with each of the FFs. The results refer to runs in arenas with hole of size $1+\varepsilon$ at distance 12 m (left) and 20 m (right) from the first VSR. Since we performed 30 replicas per experiment configuration, the maximum value obtainable is 30.

Group size	Distance 12 m			Distance 20 m		
	$\mathcal{AVG}$	$\mathcal{MAX}$	$\mathcal{MIN}$	$\mathcal{AVG}$	$\mathcal{MAX}$	$\mathcal{MIN}$
4	29	27	30	30	20	29
6	30	30	30	30	26	30
8	30	30	30	29	28	30
10	30	28	29	30	30	29

3 Results

To correctly analyze the results, we first need to reason about the meaning of the fitness computed with each FF. $\mathcal{AVG}$ considers the average x -position of the group of VSRs. This means that outliers have a small influence on the computed fitness value. A high value represents a group that overall performs well. Conversely, a low value indicates a group that overall performs poorly. Therefore, $\mathcal{AVG}$ incentivizes the VSRs to cooperate, as the average x -position of the group matters more than the individual one. Nevertheless, analyzing the collective behavior only by looking at this fitness value possibly hides some peculiar behaviors. A group that succeeds pushing a single or few individuals beyond the hole would still have a relatively low fitness, hiding this partial success. Differently, $\mathcal{MAX}$ considers the furthest x -position reached by a VSR. This means that there is not any explicit request to maximize the number VSRs beyond the hole, requiring instead just one VSR to pass it. It follows that the fitness value computed with $\mathcal{MAX}$ does not allow understanding how the group as a whole performed, again hiding some information. Finally, $\mathcal{MIN}$ only considers the leftmost VSR, which is the nearest to the start of the arena. The idea is to incentivize cooperation, as the fitness of the group depends on the position of the worst VSR. This again hides some information, as we are not aware of the fitness of any subgroup. Overall, regardless of the FF employed, we can use the right side of the hole as a coarse grain threshold for a first discrimination of the results.

The results show that the performance (*i.e.*, the fitness of the best individual at the end of the evolution) in arenas with hole of size $1+\varepsilon$ is mostly greater than the threshold, regardless of the FF employed (see Table 1). This implies that the whole group succeeds in passing the hole for $\mathcal{MIN}$, most of it pass the hole for $\mathcal{AVG}$, and at least one VSR passes the hole for $\mathcal{MAX}$. Nevertheless, the performance decreases drastically when considering the arenas with hole $2+\varepsilon$ (see Table 2). Here, just few groups obtain a performance greater than the threshold, meaning that most of the VSRs do not pass the hole. Specifically, when considering $\mathcal{MAX}$, every failure implies that no VSR succeeded in passing

Table 2. Number of replicas with a performance sufficient to be considered a “success”, with each of the FFs. The results refer to runs in arenas with hole of size $2+\varepsilon$ at distance 12 m (left) and 20 m (right) from the first VSR. Since we performed 30 replicas per experiment configuration, the maximum value obtainable is 30.

Group size	Distance 12 m			Distance 20 m		
	$\mathcal{AVG}$	$\mathcal{MAX}$	$\mathcal{MIN}$	$\mathcal{AVG}$	$\mathcal{MAX}$	$\mathcal{MIN}$
4	0	0	0	0	2	0
6	0	2	1	4	6	1
8	2	5	2	5	8	1
10	5	1	0	6	13	2

the hole. The performance decrease is even more evident in arenas with hole of size $3+\varepsilon$. In this situation, we do not register any success. The problem is likely due to the impossibility of the group to create a stable structure that can resist the gravitational force. This happens when many VSRs start to lose touch with the ground. Because of this lack of success, we decide to not show any result for arenas with hole of size $3+\varepsilon$. Overall, $\mathcal{MIN}$ seems the best for simple arenas, obtaining the same or better success rate than the other FFs while ensuring cooperation. $\mathcal{MAX}$ allows obtaining more (probably partial) successes in the hardest arenas, but performs worse than the other FFs in simpler ones. Finally, $\mathcal{AVG}$ seems to perform as well as $\mathcal{MIN}$ in the simple arenas, but performs better than it in the hardest.

Until now our analysis just considered a binary result: beyond the hole or not. However, this view is somehow limiting, as it does not describe how well the groups performs after passing the hole (*i.e.*, how far they reach). Figure 2 shows the distribution of the performance for each FFs, group size, and hole distance, in arenas with hole of size $1+\varepsilon$. As visible, the maximum performances are more or less the same regardless of the FF employed. Nevertheless, the overall distribution changes according to the measure we use. Specifically, $\mathcal{AVG}$ and $\mathcal{MIN}$ present very compact distributions near the maximum. This indicates that the performance is stable across different groups. Nevertheless, this is not the case for $\mathcal{MAX}$. The distributions obtained employing this FF are indeed much wider, indicating that the results are less stable. When considering arenas with holes of size $2+\varepsilon$, we see instead that $\mathcal{MIN}$ produces solutions that perform very poorly (see Fig. 3). Differently, $\mathcal{AVG}$ and $\mathcal{MAX}$ still succeed in passing the hole, but the difference in distributions between the two decreases.

As anticipated at the beginning of this section, the performance does not give us full information on the ability to pass the hole. Therefore, we consider the final x -position of all the VSRs of the group to understand how many passed it. We normalize the results dividing the number of VSRs beyond the hole by their total number in the experiment configuration. Figure 4 shows the results in arenas with hole of size $1+\varepsilon$. Since $\mathcal{MIN}$ requires the VSRs to cross altogether, the results tend to be polarized: they succeed, or they do not. Solutions which

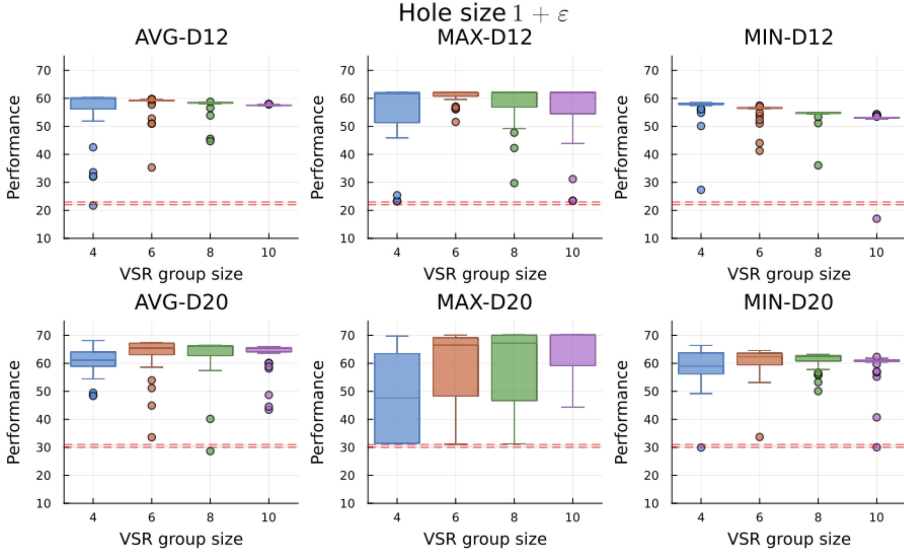


Fig. 2. Performance distribution in the arenas with hole size $1 + \varepsilon$. Each distribution, one per experiment configuration, consists of 30 values. The configurations of the experiment differ by the FF used, the distance of the hole from the starting x -position of the VSRs, and the group size. The plots on the first, second, and third column respectively represent experiments run with FFs AVG , MAX , and MIN . The plots on the first and second row respectively represent experiments with the hole at distance 12 m and 20 m from the x -position of the first VSR. The red band in the plots represents the hole.

allow just some VSRs to pass are completely ignored during evolution, and thus do not often appear in the final results. AVG presents similar results, but being less strict it does not discard solutions which leave some VSRs behind. This results in some of the runs having just part of the group beyond the hole. For MAX the ratio of VSRs left behind increases significantly, especially in the arenas with the hole at 20 m distance. Nevertheless, most of the results still show groups larger than one VSR passing the hole, indicating that the evolution still somehow selects cooperation. When considering arenas with hole of size $2 + \varepsilon$, the pass-ratio of MIN drops almost always to zero (see Fig. 5). This is again due to the polarization induced by this FF. Even the passing ratios of AVG and MAX decrease, becoming more similar to each other. These results confirm the previous considerations on the effectiveness of the different FFs, indicating MIN as the best for simple arenas, MAX as the best for hardest ones, and AVG as an effective trade-off between the other two.

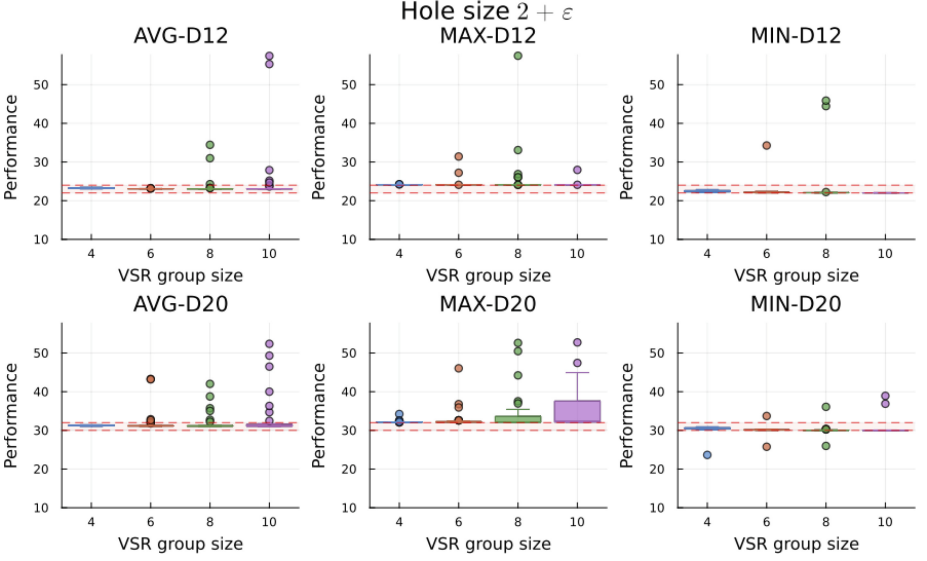


Fig. 3. Performance distribution in the arenas with hole size $2 + \varepsilon$. Each distribution, one per experiment configuration, consists of 30 values. The configurations of the experiment differ by the FF used, the distance of the hole from the starting x -position of the VSRs, and the group size. The plots on the first, second, and third column respectively represent experiments run with FFs $\mathcal{AVG}$, $\mathcal{MAX}$, and $\mathcal{MIN}$. The plots on the first and second row respectively represent experiments with the hole at distance 12m and 20m from the x -position of the first VSR. The red band in the plots represents the hole. (Color figure online)

Another important aspect to investigate is the impact of the number of VSRs on the performance and passing ratio. Starting from the former, we recognize that the size of the group has a different impact according to the considered FF (see Figs. 2 and 3). Increasing the VSRs in $\mathcal{AVG}$ seems to increase the performance of the group. In arenas with hole of size $1 + \varepsilon$, a difference is clear mostly for the smaller groups of size 4. In arenas with hole of size $2 + \varepsilon$, the difference seems more evident across all the groups. For $\mathcal{MAX}$, the increase in performance is evident in arenas with both the hole sizes. Specifically, increasing the size of the group seems to affect the mean and spread of the distributions significantly. For $\mathcal{MIN}$ we witness instead a different trend. In the arenas with the smaller hole size, the performance seems to decrease at the increase in the group size. Differently, results obtained in arenas with hole of size $2 + \varepsilon$ do not show any noticeable difference across different group sizes. These peculiar trends are due to the fact that having more VSRs requires a higher number of individuals passing the hole. This obviously complicates the task, as with $\mathcal{MIN}$ a single error affects the fitness of the whole group.

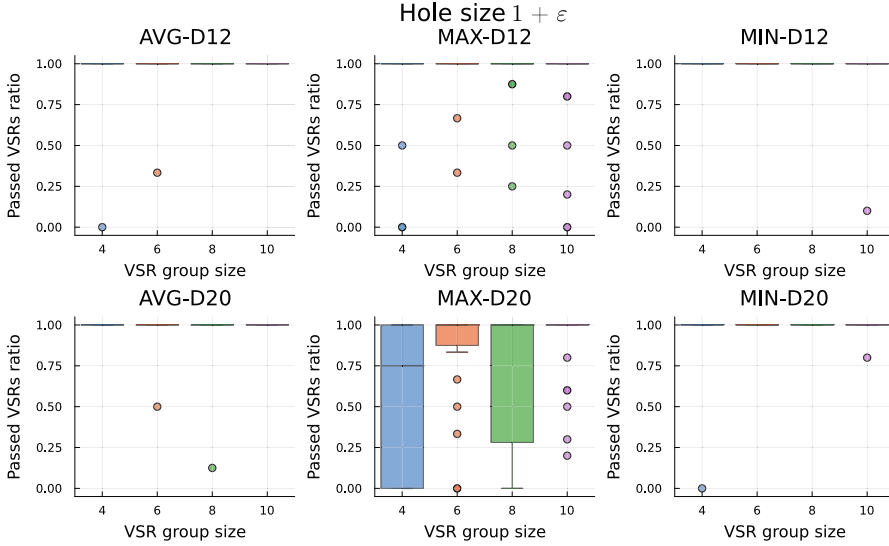


Fig. 4. Ratio distribution of VSRs which have passed the hole of size $1 + \varepsilon$. For each of the 30 replicas of experiment configurations, we calculate the number of VSRs beyond the hole and divide it by the group size. A value of 1 means that all the VSRs passed the hole. A value of 0 means that no VSR passed the hole. Each distribution, one per experiment configuration, consists of 30 values. The configurations of the experiment differ by the FF used, the distance of the hole from the starting x -position of the VSRs, and the group size. The plots on the first, second, and third column respectively represent experiments run with FFs *AVG*, *MAX*, and *MIN*. The plots on the first and second row respectively represent experiments with the hole at distance 12 m and 20 m from the x -position of the first VSR.

We witness similar results when considering the passing ratio of a group. For *AVG* the metric increases slightly when increasing the group size in the arenas with hole of size $2 + \varepsilon$ at distance 20 m. For *MAX* we recognize an increment in arenas with both the hole sizes. This seems to confirm that *MAX* is more sensible to the group size than the other two FFs. For *MIN* we do not witness any particular difference.

Overall, both the results on the pure performance and the passing ratio show that increasing the group size leads to an improvement. The degree of the improvement strongly depends on the FF employed.

4 Discussion

In this work we demonstrate the feasibility of self-assembly in a group of VSRs to pass over a hole in the ground. We achieve this by evolving the VSR controller by means of an EA. Three different FFs guide the evolutionary process, namely: *AVG*, *MAX*, *MIN*. We find that the performance of the group depends on

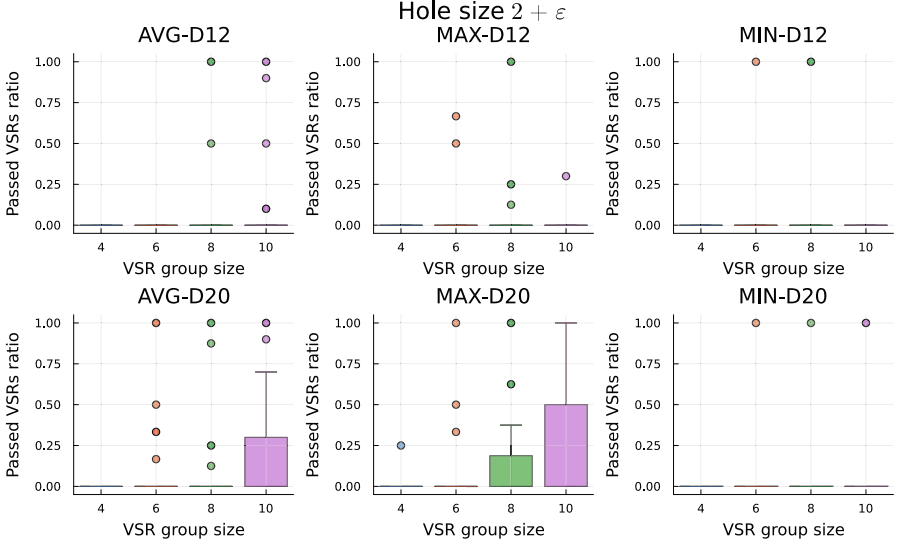


Fig. 5. Ratio distribution of VSRs which have passed the hole of size $2 + \varepsilon$. For each of the 30 replicas of experiment configurations, we calculate the number of VSRs beyond the hole and divide it by the group size. A value of 1 means that all the VSRs passed the hole. A value of 0 means that no VSR passed the hole. Each distribution, one per experiment configuration, consists of 30 values. The configurations of the experiment differ by the FF used, the distance of the hole from the starting x -position of the VSRs, and the group size. The plots on the first, second, and third column respectively represent experiments run with FFs *AVG*, *MAX*, and *MIN*. The plots on the first and second row respectively represent experiments with the hole at distance 12m and 20m from the x -position of the first VSR.

the FF employed. Specifically, *MIN* performs well when the arena is simple, but it is very sensible to failures. As this metric considers the x -position of the leftmost individual, a single VSR falling in the hole is enough to hinder the fitness of all the group. This evaluation approach is especially strict, as it ignores the difference between an entire group falling in the hole and just some of its members. The result is that partial successes that could be further improved by the EA are discarded, and every epoch of the evolution results in a sort of random search. It follows that *MIN* is enough to solve easy problems, but becomes ineffective as soon as the complexity of the task increases and refinements of the solutions become necessary. *MAX* follows a diametrically opposite approach by considering the x -position of the rightmost individual. This feedback is more easily exploited by the EA during the evolutionary search. Indeed, a controller succeeding in passing at least one VSR is forwarded and used in the generation of the next population. Nevertheless, this FF still presents some issues, as maximizing the passing ratio only happens if it gives advantage beyond the hole. For instance, having a larger group on the right side of the hole can enable a more

effective motion, increasing the fitness and thus pushing for an increased passing ratio. Overall, we notice that many controllers produced with $\mathcal{MAX}$ leave a significant amount of VSRs behind. This may indicate that increasing the group size beyond the hole does not necessarily improve the motion, or that this further refinement is too complex to achieve. Finally, $\mathcal{AVG}$ represents a combination of the two previous FFs, trying to provide a more fine-grained evaluation of the controllers. This considers indeed the average x -position of all the VSR, overcoming the problem of considering only the best or the worst individual. The expected outcome is the maximization of the size and speed of the group beyond the hole, as it would have evolutionary advantage against single individuals and slow groups. Overall, we identify a successful trend induced by the use of $\mathcal{AVG}$. This is however comparable to that of $\mathcal{MAX}$ in arenas with hole of size $2 + \varepsilon$, rising the question on whether the finer evaluation of $\mathcal{AVG}$ is really beneficial. We believe that despite similar results, $\mathcal{AVG}$ is still worth using as it provides better and most stable results in arenas with hole of size $1 + \varepsilon$, and can potentially be more effective when the size of the group grows largely. Overall, we believe that the best FFs are $\mathcal{AVG}$ and $\mathcal{MAX}$, depending on nuances of our goal. $\mathcal{MIN}$ seems instead not effective enough, being too sensible to single failures. Finally, as the FFs induce different search behaviors, they might be combined, each to be used in different phases of the search.

Another factor that possibly affects the success of the group of VSRs is their initial distance from the hole. Indeed, self-assembling requires some space to operate. In this work we test two different distances in order to assess if they lead to any change in the results. Specifically, we test with holes 12 m and 20 m distant from the starting position of the VSRs group. We find some differences in the results. Increasing the distance of the hole overall improves the performance. We believe this is due to the fact that the group of VSRs has more space to create a stable structure, synchronize the movement, and acquire momentum to pass the hole. Nevertheless, we find an exception when using $\mathcal{MAX}$ in arenas with hole of size $1 + \varepsilon$. In this scenario, the performance obtained decreases with the increase of the distance. The motivation behind this different behavior is currently unclear.

We also investigate if the size of the VSRs group affects the results. We identify a positive correlation between the group size and its effectiveness in the task. This means that increasing the number of individuals in the group seems to generally improve the performance. The motivation is probably that a larger group is more stable, allowing to withstand the gravitational force dragging part of it into the hole. Additionally, a larger group can possibly employ more advanced motion strategies, moving further beyond the hole. This general considerations do not apply however to $\mathcal{MIN}$, which display negligible and sometimes even decreasing performance with the increase in size. We believe this is due to the fact that increasing the number of individuals increases the probability of fails, affecting the worst-based fitness calculation of $\mathcal{MIN}$.

One interesting aspect that we observed in some preliminary results, but that was not replicated in the final ones, is that soft-robotics enables strategies that

are not possible with stiff robots. More specifically, we saw the group of VSRs finding a solution for the arena with the hole of size $3+\varepsilon$. The VSRs form a chain which heads directly towards the bottom of the hole. By maintaining the chain rigid, the tail gains centripetal acceleration (as it happens in a catapult). When the tail of the chain perceives this sudden acceleration, it detaches, being thrown on the other side of the hole. This peculiar strategy is in fact only possible if we consider using soft-robots, which can withstand being tossed on the ground. This is obviously not the focus of the work, and it was neither replicated in the final results, nevertheless, we consider it interesting and a clear example of what soft-robotics enables.

5 Conclusion

In this work we assess if it is possible to evolve the controller of VSRs to self-assemble and solve a hole-passing task. To verify the scalability of the approach, we experiment with multiple arenas of increasing complexity and with groups of different size. Additionally, we investigate how different FFs affect the performance of the group.

Our results show that it is possible to generate a controller that induces self-assembly and allows passing the hole. The performance varies according to the complexity of the arena, ranging from almost 100% of success in the simplest arena to 0% in the most complex one. Specifically, we note that increasing the distance between the hole and the group of VSRs generally increases the performance. We attribute this effect to the possibility of the group to form more stable and effective structures. We find that the performance of the group also depends on the FF employed. Specifically, evaluating the group according to its worst individual (*i.e.*, $\mathcal{MIN}$) generates underperforming groups in complex arenas. Conversely, using the best individual (*i.e.*, $\mathcal{MAX}$) produces good results, but introduces some fluctuations. The best FF overall seems to be $\mathcal{AVG}$, which produces more stable, but still good, results. Finally, we identify scalability in the approach, with the group performance usually increasing with its size. This does not happen however when employing the $\mathcal{MIN}$ FF. In this case, the performance seems even negatively affected by the increase in group size. This is probably due to the fact that increasing the number of individuals increases the probability of fails, affecting the worst-based fitness calculation of $\mathcal{MIN}$.

We intend to perform additional experiments to verify if a pre-training or a scaffolding may lead to better results in the case of large holes. Indeed, the EA starts the evolution in each arena without any previous knowledge. This requires rediscovering fundamental behaviors every time. We expect that evolving VSRs able to solve simpler arenas will speed up the process and increase the percentage of success.

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